Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/04/2025
PI Name:	Ditzler, Mark A. Ph.D.		
Project Title:	Evolutionary Processes That Drove the Emergence and Early Distribution of Life (EPDEL)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	
Organization Name:	NASA Ames Research Center		
PI Address 1:	Space Science and Astrobiology Division (ST)		
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Zip Code:	94035	**Congressional District:**	16
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2022	**End Date:**	09/30/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Note: Per the Principal Investigator (PI), both Andrew Pohorille and Timothy Lee were part of the original research team; however, the PI reports that they have passed away (Ed., 6/4/25).		
COI Name (Institution):	Rothschild , Lynn (NASA) Cooper, George (NASA) Almeida, Eduardo (NASA) Nuevo, Michel (NASA) Popović, Milena (BMSIS) Rios, Andro (BMSIS) Zajkowski, Tomasz (BMSIS) Wilson, Michael (SETI) Wei, Chenyu (SETI) Bera, Partha (BAERI)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose continuation of a long-term program of research and service to the external astrobiology community that aims at an integrated approach to investigating the origin and early evolution of life. Specifically, we focus on two key features of the emergence of life: 1) continuity between and within chemical and biological evolution, and 2) the integration of multiple systems in the emergence and early proliferation of life. This activity supports NASA's goal of understanding the origin and early evolution of life on Earth and determining the potential for life to emerge in other habitable environments. This activity consists of four research tasks: A) "From abiotic formation of amphiphiles to protocellular function", B) "Pyruvate reaction networks and the origin of metabolism", C) "Evolution of functional peptides and protocell adaptation", and D) "Tools for applying systems chemistry/biology to protocellular processes", along with service activities that support the larger astrobiology community.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In our project, we have initiated a long-term program of research and service to the external astrobiology community that aims at an integrated approach to investigating the origin and early evolution of life. Specifically, we focus on two key features of the emergence of life: • Continuity between and within chemical and biological evolution. • The integration of multiple systems in the emergence and early proliferation of life. (I) Continuity between and within chemical and biological evolution. Raw materials from which protocellular ecosystems were formed were derived from astrochemical and geological processes. There must have been continuous paths between these processes and biological evolution on Earth and potentially elsewhere in the universe. Demonstrating production of molecules present in modern biology through abiotic processes is not sufficient to connect abiotic to biotic processes. Instead, we aim to establish plausible continuous paths from abiotic processes in astrochemistry and terrestrial chemistry to the emergence of protocellular compartments and metabolic processes that could support survival, reproduction, and open-ended evolution in protocellular ecosystems. (II) The integration of multiple systems in the emergence and early proliferation of life. In emerging life, protocells must evolve as a system. In these evolving entities, all components work in concert. This creates an imperative for coevolution of different functions that involves, among others, the formation and maintenance of semipermeable cell boundaries, simple metabolism, and self-organization of molecules to form functional structures. This means that evolution of each component of a protocellular system was both promoted and constrained by evolution of other components. We aim to understand how collective evolution, a critically important but still poorly understood process, played out during the emergence of life. EPDEL addresses NASA's goals of understanding the origin and early evolution of life on Earth and determining the potential for life to emerge in other habitable environments. Specifically, it will markedly improve our understanding of "chemical systems that could have served as precursors of metabolic and replicating systems on Earth and elsewhere, including alternatives to the current DNA-RNA-protein basis for life". Tasks A and B will improve an understanding of "the mechanisms that led to the production of prebiotic molecules in various environments" which is "critical for establishing the inventory of ingredients from which life originated on Earth", and Task C will help to "characterize the overall chemical underpinnings of the processes that lead to the persistence and function of evolvable macromolecular systems".
Bibliography Type:	Description: (Last Updated: 11/17/2025)
Articles in Peer-reviewed Journals	Carter CW Jr, Tang GQ, Patra SK, Dieckhaus H, Kuhlman B, Douglas J, Wills PR, Bouckaert R, Popovic M, Ditzler MA. "Structural enzymology, phylogenetics, differentiation, and symbolic reflexivity at the dawn of biology." Genome Biol Evol. 2025 June;17(6):evaf095. https://doi.org/10.1093/gbe/evaf095 ; PubMed PMID: 40476352; PubMed Central PMCID: PMC12152416 , Jun-2025